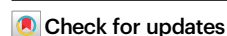


Individual locomotor bias drives counterclockwise motion in pedestrian crowds

Received: 29 April 2025

Accepted: 14 May 2026

Published online: 10 June 2026



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Pedestrian walking behaviour is intrinsic to individuals, yet it is influenced by external factors such as obstacles and the degree of crowding. It is precisely in crowded scenarios that pedestrian interactions lead to collective motions, such as lane formation or waves. Recently, the spontaneous development of collective counterclockwise motion has been reported in both dense and sparse human assemblies. Here we present five experimental studies of this phenomenon conducted across diverse conditions in Spain and Japan, demonstrating that counterclockwise bias in roaming pedestrians is a robust and reproducible feature and originates from individual tendencies rather than from collective interactions. These findings challenge the traditional view that social dynamics shape pedestrian motion, highlighting the existence of an intrinsic locomotor bias.

Anyone who has walked along a crowded street has probably noticed that pedestrians spontaneously self-organise into lanes, grouping with pairs that share the same direction of motion^{1–3}. This phenomenon can be seen as an example of the crowd inducing individual behavioural changes that result in an overall benefit for the group, as lane formation reduces personal discomfort and minimises the risk of collision⁴. Another scenario in which collective patterns emerge is when a crowd exits through a narrow passage—so small that two people cannot pass through it simultaneously⁵. In this scenario, pedestrians naturally split into two alternating streams—one passing through the door's right-hand side and the other through its left-hand side. This so-called zipper effect results in a more efficient evacuation than if they had simply formed a single file and exited the room through the centre of the door^{5,6}.

Interestingly, in both cases (cross-flow and bottleneck flow) self-organised structures form owing to individual collision-avoidance manoeuvres and an unspoken mutual communication between people⁷. In other words, a simple individual behaviour adopted

independently by many people can result in a collective behaviour that is only indirectly the outcome of each individual's action. This is what is called an emerging phenomenon, with examples in pedestrian dynamics extending to group oscillations^{8,9}, stripes^{10,11}, and waves¹² observed in large, dense crowds. Remarkably, all these phenomena occur without any leader orchestrating them, and people are often not even aware of the pattern they are creating.

Furthermore, it has been argued that, under some circumstances, a seemingly collective pattern is created (or strongly influenced) by biased preferences of the members of a crowd. For example, in most countries, the lanes described earlier tend to form to the right (in the sense of the march) as a result of a weak tendency for people to move rightwards when facing another pedestrian¹³. Similarly, it has recently been proposed that a slight preference among right handed people to turn left when facing a wall¹⁴ could underlie the emergence of collective counterclockwise (CCW) crowd motion, both in mosh pit dancing¹⁵ and when a crowd walks freely within an arena¹⁶.

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In contrast to this view, our study offers a different perspective on the origins of CCW motion. Through five carefully designed experimental campaigns conducted in diverse settings and across different countries, we have gathered evidence that challenges the conventional interpretation. Rather than being an emergent property driven by interpersonal interactions (possibly influenced by personal biases), our results indicate that the collective CCW motion is rooted in inherent individual tendencies. As we shall see, we observed that CCW motion consistently emerged even when all pedestrians roaming in an enclosed space were left-handed or when their turning preference was to the right (Fig. 1a). We also ruled out the possibility that the cause is associated with interactions with boundaries by conducting experiments in an open space (Fig. 1b). Another plausible hypothesis related to interpersonal interactions—in particular, to avoidance manoeuvres—suggested that such manoeuvres might trigger CCW rotation in the same way as they lead to right-side lane formation in counterflows. However, the results from experiments in Japan (Fig. 1c), where lanes tend to form to the left side (in the sense of the march) during counterflows, refuted this idea. Moreover, we excluded a strong effect of social or acquired influences (such as the CCW sense of motion in athletics tracks) by analysing the dynamics of children during free play at a Japanese nursery¹⁷. We also found no evidence that an unspoken social norm is responsible for CCW motion by showing that the survey responses do not reveal a clear norm. Finally, we analysed single pedestrians walking alone in an enclosure, confirming that this symmetry-breaking phenomenon is caused by individual behaviour, most likely biologically rooted.

Our contribution is thus twofold. First, we provide experimental evidence that the CCW bias is robust across diverse experimental settings and is reproducible across the two countries represented in our samples (Spain and Japan), with potential implications for urban planning and crowd management. Second, our findings demonstrate that this phenomenon arises from individual behaviour rather than collectively emerging due to pedestrian-pedestrian or pedestrian-boundary interactions. Then, we ruled out some of the most obvious individual symmetry breaking factors—such as handedness, footedness, and eye dominance—thus leaving the precise origins of this intriguing behaviour open for further investigation.

Results

In this section, we present our results on the statistical properties of motion observed in each experiment. By analysing the patterns and

differences across scenarios and countries, we aim to uncover the underlying mechanisms driving the consistent CCW asymmetry in human motion and offer explanations for its prevalence.

Confined random motion in Spain

Our initial study was carried out in Spain with the aim of corroborating that CCW motion is caused by a small bias in the turning preference of pedestrians when facing a wall (right-handed people prefer turning towards the left¹⁴). Under this premise, we implemented experiments in which groups of people with different handedness and turning preferences were asked to roam a 5 metres radius circular arena (Fig. 1a). The turning preference of each participant was identified before the group trials. To this end, each volunteer was instructed to walk along a straight line until reaching a wall, execute a 180-degree turn, and return. In this way, individuals were categorised as either Right-Turners (RT) or Left-Turners (LT) depending on their turning direction. Independently, volunteers who were both left-handed and left-footed were categorised as Left-Dominant (LD).

In each experiment, participants moved freely within the arena for three 40-s intervals interspersed with two phases in which they were asked to navigate to a designated point. These distinct movement phases are clearly identifiable by analysing the temporal evolution of the average speed of the group (inset of Fig. 2, upper panel). Free movement periods (highlighted in colour) display a notably higher speed than the beginning of the experiment or the directed movement phases (in grey). To quantify the directionality of rotation, we employed the polarization parameter M defined at each time step as the average of the individual polarizations $m_i(t)$. The latter are computed as $m_i(t) = \hat{v}_i(t) \cdot \hat{e}_i^\phi(t)$, where $\hat{v}_i(t)$ is the normalized velocity vector of pedestrian i and $\hat{e}_i^\phi(t)$ is the azimuthal unit vector relative to a central point¹⁸. A complete description of this measure and its variants is provided in the “Methods”. An example of the temporal evolution of $M(t)$ is depicted in the inset of Fig. 2 (lower panel). To quantify the system's net rotational tendency, we computed the time-averaged polarization $\bar{M}$ during each interval of free motion. $\bar{M} > 0$ corresponds to CCW motion, whereas $\bar{M} < 0$ indicates clockwise (CW) motion.

As explained, it was expected that increasing the proportion of right-turners in the experiment would favour CW rotation. But the results revealed that neither the number of participants nor the proportion of right-turners significantly influenced $\bar{M}$. Instead, across all experimental conditions, $\bar{M}$ consistently exhibited a positive value around $\bar{M} \sim 0.2$ (Fig. 1d, orange), indicating a robust and persistent

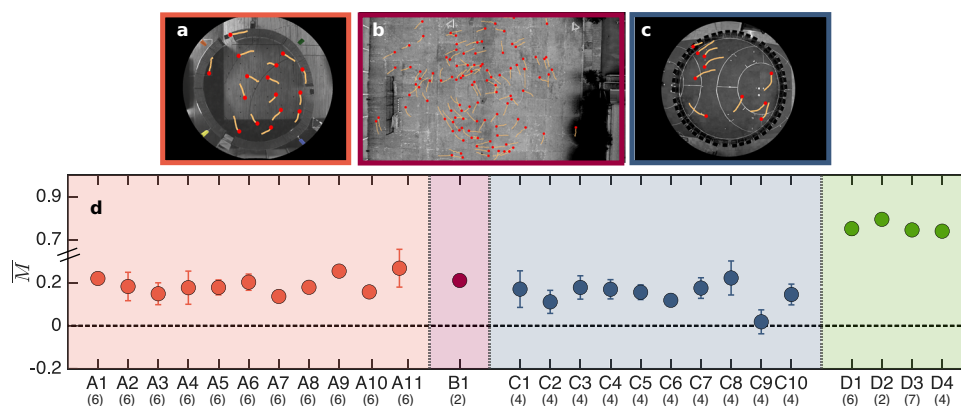


Fig. 1 | Experimental setups and collective turning polarization. Panels (a–c) show snapshots of the different analysed experimental scenarios, illustrating pedestrian trajectories over the last 2 s (orange) and their current positions (red). For image rights purposes, individual pedestrians have been removed from these snapshots; only the static arena background and their superimposed trajectories are displayed. **a** Confined random motion in Spain, **b** random motion of Spanish

teenagers in a schoolyard, **c** confined random motion in Japan. Panel (d) shows the time-averaged collective polarization ($\bar{M}$, as defined in “Methods”) for the different experimental conditions evaluated in each scenario (see Supplementary Table 1 for more details). Error bars indicate the standard error of the mean. Sample sizes used to compute $\bar{M}$ and the standard error of the mean are reported in parentheses below each experimental condition.

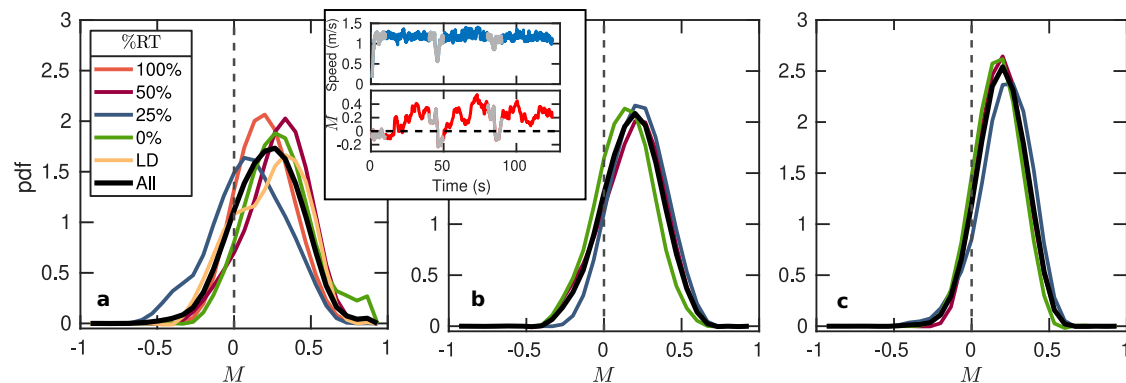


Fig. 2 | Collective rotational behaviour on confined random motion in Spain. The panels show the probability density functions (PDFs) of the collective polarization values (M), for groups with different numbers of participants: **a** 16, **b** 24, and **c** 32. Different colours (see legend) are used for crowds with different percentages of right-turners (%RT) and for the case with only Left-Dominant (LD) pedestrians. The black line represents the aggregated distribution obtained by combining data

from all experimental conditions. Inset: time series of the average speed of all participants (top) and the collective polarization (bottom). The values used to generate the PDFs are the ones marked on red. Intervals covering the initial stage of the experiment and periods of directed motion towards the walls (grey in the inset), were identified by analysing the average speed, and excluded from the analysis.

CCW bias. In this sense, it is noteworthy that even experiments A1 and A11 (in which 100% of pedestrians were right-turners and left-handed, respectively) revealed a similar, positive value of $\bar{M}$.

To further understand this observation, we analysed the probability density functions of M (Fig. 2). Interestingly, regardless of the global density (increasing from **a** to **c**) and the proportion of right-turners, all distributions are shifted towards positive values and are unimodal, with the peaks centred at $M \sim 0.25$. The low proportion of $M < 0$ values implies that the system maintains a constant CCW rotation. At the same time, the absence of values at $M \sim 1$ indicates that the CCW motion is not a global effect involving all pedestrians. Interestingly, the distributions become narrower as the number of pedestrians increases, which may suggest the existence of a collective effect that boosts the stability and robustness of the CCW rotation. More importantly, the overlap of all distributions obtained with the same number of pedestrians but different proportions of right-turners (especially for large crowds) indicates that individual turning preferences have a negligible impact on the emergence of CCW behaviour.

Next, aiming to elucidate the actual role of boundaries in the development of CCW motion, we analysed the spatial distributions of density, velocity, and polarization within the arena (Fig. 3). The density fields (first row, Fig. 3) show a rather homogeneous spatial distribution, yet some faint circular patterns can be perceived. These suggest that the position of the boundaries (or pedestrians' perception of them) affects the motion within the arena. The velocity fields (second row, Fig. 3) reveal that the CCW motion extends over the whole arena but is slightly more pronounced near the boundaries. This is further confirmed by examining the polarization fields (third row, Fig. 3). On average, regions coded in blue are more abundant over the whole arena, but the colours are more intense near the boundaries; hence suggesting a possible role of those in the development of CCW rotation.

Boundary-free experiment in a schoolyard

From previous results, and in order to clarify whether the boundaries really trigger the CCW motion or just help to stabilise (and perhaps magnify) it, we designed a follow-up experiment in which pedestrians walked in an open and practically unconstrained setting (Fig. 1b). This consisted of a $50 \times 60 \text{ m}^2$ schoolyard in Spain, where over one hundred teenage students were gathered (see “Methods” for details). Surprisingly, despite the influence of boundaries being practically suppressed, the CCW rotation persisted, as reflected by the positive value of $\bar{M}$ depicted in Fig. 1d (red). In agreement with this, the analysis of the

PDF(M) reveals again a unimodal distribution shifted towards positive values (Fig. 4a). Interestingly, the PDF(M) is even narrower than the ones presented in Fig. 2, suggesting that the variable controlling the width of the distribution is the total number of pedestrians, not the density—which in this case is 6 times lower than in the sparser experiments of the first scenario.

Confined random motion in Japan

After ruling out the pedestrian-boundary interactions as the origin of the CCW rotation, we focused on pedestrian-pedestrian interactions as a potential driving mechanism. Pedestrian-pedestrian interactions are key to various self-organising behaviours, with lane formation in bidirectional flows being a prime example^{4,19,20}. This process arises from local coordination, where individuals adjust their paths during head-on encounters to avoid collisions²¹. In Spain (and most European countries), this avoidance manoeuvre is typically implemented by moving towards the right-hand side^{13,22}, hence leading to the symmetry breaking in lane formation. After this fact, the hypothesis was that if pedestrians avoid collisions by stepping to the right side (thus leaving the incoming person to the left), in the circular arena they would end up moving CCW near the boundary walls.

To test this idea, we conducted experiments in Japan, a country where lanes in bidirectional flows conspicuously appear on the left side, as pedestrians generally avoid others by stepping to the left. First, we confirmed this left-side stepping tendency through a questionnaire where participants indicated their natural avoidance direction when viewing corridor walking images (see Supplementary Note 1 for details). We then performed new experiments in an enclosure similar to the one used in Spain (Fig. 1c), following the same methodology. Unexpectedly, the positive values of M reported in Fig. 1d reveal that the CCW motion persisted, hence refuting the idea that the stepping aside pedestrian manoeuvres were behind the collective development of CCW motion. Indeed, $\bar{M} > 0$ in all experimental trials but one (C9 in which $\bar{M} \approx 0$), an exception that we attribute to the intrinsic variability of human behaviour. Furthermore, as already observed in experiments 1 and 2, the distributions of M (Fig. 4b, c) remain skewed towards positive values, and the peak (at about $M \sim 0.2$) is more pronounced as the number of pedestrians in the arena increases; i.e., the fluctuations of M are smaller as the population size grows.

Random motion in a nursery school

We then addressed the question of whether social rules or learned behaviours—potentially shaped by sporting events like athletics or

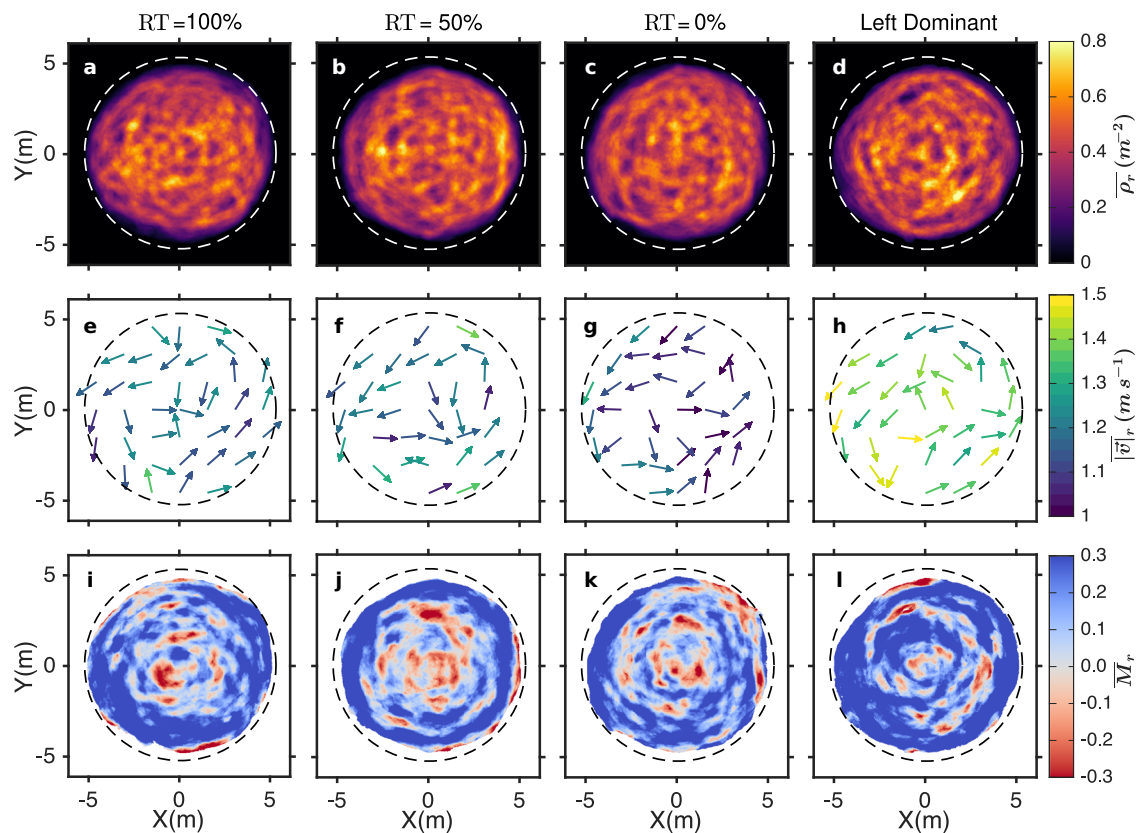


Fig. 3 | Density, velocity and polarization fields for varying proportions of right-turners in the Spanish confined scenario. Spatial distribution of temporally averaged density $\bar{\rho}_r$ (first row), velocity $\bar{v}_r$ (second row), and polarization $\bar{M}_r$ (third row) fields for a crowd of 16 pedestrians with different turning preferences as indicated at the top. The colour scales on the right (same for all cases) indicate (i)

the average local density in persons/m² (a–d); (ii) the average speed in m/s (e–h); and (iii) the average local polarization value (i–l). In (e–h), the arrows indicate the average direction of the local velocity vector. The spatial units in both the vertical and horizontal directions are metres for all plots.

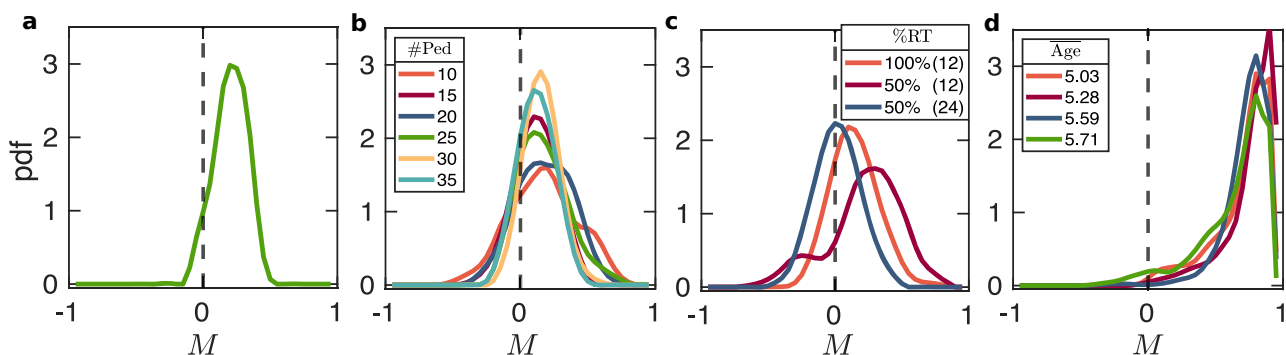


Fig. 4 | Probability density function (PDF) of collective polarization values (M) in different scenarios. In (a) the boundary-free motion of teenagers in Spain. In b, c the confined motion in Japan. In d the children's motion in a Japanese nursery school. In each panel, colours are used to label different experimental conditions,

as described in the legends. In a, only one experimental condition is considered. In b, each curve corresponds to a different crowd size. In c, both the percentage of right-turners (%RT) and the group size (12, 24) varies. In d, four realizations with different children of slightly different age are reported.

other learned behavioural habits—might be the cause of the CCW collective motion. To explore this option, we analysed experiments conducted in a nursery school, as previous research has demonstrated that young children differ from adults in their emergent movement patterns²³, and it can reasonably be assumed that they are less influenced by acquired adult conventions (e.g. signage norms or circulation habits). In these experiments, conducted by Ichikawa et al.¹⁷, children (about 5 years old) participated in an eurhythmics activity involving free running (see “Methods” for more details). Interestingly, the CCW

motion not only develops as in previous scenarios, but it becomes much more pronounced, as revealed by the higher values of $\bar{M}$, systematically above $\bar{M} = 0.7$ (Fig. 1d, green). This behaviour is corroborated by the distributions of M (Fig. 4d), which show a noticeable peak near $M \sim 1$ that indicates a highly consistent and stable vortex-like motion, with all children moving in unison. This suggests that children, at least in this specific activity, tend to imitate their peers and end up walking in the same direction, which is, of course, the CCW one. This is consistent with prior research showing that young children are highly

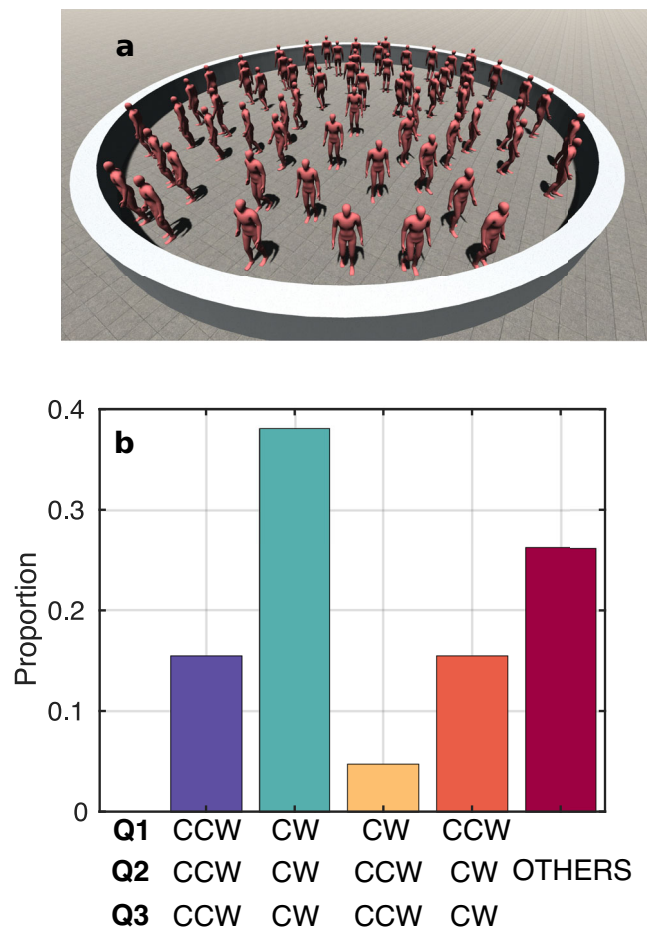


Fig. 5 | Influence of social norms. **a** Photograph used in the survey, where participants responded to three questions (Q1, Q2, Q3) about the direction of rotation they would choose. See *Methods* for the complete survey form. **b** Proportion of responses to the survey questions. Answers, limited to CW or CCW, are grouped into three categories: (i) The same answer for all three questions (all CW or all CCW), indicating a strong influence of social norms, (ii) The same answers for Q2 and Q3 but different from Q1, suggesting a moderate influence of social norms, and (iii) Mixed answers, reflecting a weak influence of social norms.

sensitive to peer consensus, prone to over-imitation, and tend to align more strongly with others during rhythmic or musical group activities^{24–27}.

Social norm elicitation

In pedestrian dynamics, it is known that social influence is behind the observed behaviour in many different instances (for examples on the social influence or the emergence of social norms, see refs. 28–30 for non-emergencies and³¹ for emergency scenarios). Therefore, to investigate the possibility that unknown social norms were behind the emergence of CCW motion, we used the notion of social norm introduced by Bicchieri³², which arises from the consideration of the expectations of people about a given situation. Two kinds of expectations are taken into account. Empirical expectations (often referred to as descriptive norms³³) correspond to what individuals think others in their reference group will do when faced with the situation of interest. Normative expectations (also called injunctive norms) refer to what individuals think the rest of their reference group expects them to do. Normative expectations are generally accompanied by the assumption that if individuals do not conform to the expected behaviour, they will be sanctioned or punished in a number of different ways. In this framework, we say that a social norm in a group exists if the majority of people in the group share common empirical and

normative expectations, and the two types agree on the behaviour to be followed. Expectations are then elicited by means of a questionnaire³⁴. In our case, this test was composed of three different questions which allowed us to identify the personal beliefs (Q1), the empirical expectation (Q2), and the normative expectation (Q3) of participants (see *Methods*). This survey was performed in Spain with a group of 168 participants.

The results of this study are presented in Fig. 5. Panel (a) illustrates the hypothetical scenario shown to the survey respondents, while panel (b) summarizes their responses. As can be seen, if a social norm is indeed present, it would surprisingly be to move CW: nearly 40% of respondents exhibited aligned empirical and normative expectations in the CW direction (i.e., Q2 and Q3 both indicated CW) and also reported a personal inclination to move CW (Q1), while another 15% of the participants also shared those expectations even if they would move in the CCW direction. This must be compared to roughly 20% of respondents who expect to move CCW, while approximately 25% provided conflicting answers. Therefore, we must conclude that a clear norm does not exist, but in case we would accept slightly more than the majority's expectations as a norm, it surprisingly would go against the observed behaviour.

The above result has interesting cognitive implications. In most social interaction contexts, people follow explicit and implicit norms that tell them how to behave in those situations, typically coming from moral, ethical, legal regulations, conventions or shared social norms. Explicit norms are a direct result of codified rules (e.g., laws or regulations), whereas implicit norms are situation-specific and act without conscious awareness. The norm in place emerges through interactions between people and the environment, potentially resulting in a situation where there is a contrast between the norm eventually adopted by people and what is expected through explicit rules. Our experiment shows that turning direction is likely not determined by explicit rules, as confirmed through the results from the questionnaire contrasting what is expected from codified motion rules. One could then conclude that the CCW turning norm (if there is any) is more likely implicit, consequently acting with participants not being aware of it. However, although the explicit vs. implicit framework allows to examine the observed CCW turning behaviour from a more systematic perspective, the mechanisms leading to this specific norm (or behavioural repertoire, to use an alternative expression) are still unclear and would deserve an in-depth investigation.

Individual behaviour

Thus far, the analysis of collective polarization M across different experiments has demonstrated the consistency and robustness of the CCW rotation effect. Moreover, the distributions of M revealed that this effect persists over time, with fluctuations around the average being dependent on the total number of pedestrians and not on the density of them. More importantly, the absence of values at $M = 1$ in all the systems but in the Japanese nursery school, indicates that the CCW motion is not a global effect involving all pedestrians. This seems reasonable, as pedestrian behaviour exhibits inherent variability and, although on average the collectivity is always rotating CCW, there might be individuals moving in the opposite direction.

To quantify this, we took advantage of our experimental capabilities, which enable precise tracking of each pedestrian and analysed individual behaviour using the individual polarization parameter m_i . Unlike M , which captures collective motion, m_i quantifies each pedestrian rotation pattern, providing insight into the individual behaviour. As an example, in Fig. 6a–d we illustrate four typical trajectories from the Japan experiment together with their corresponding PDF(m_i) (Fig. 6e–h). Figure 6a displays a stable CCW trajectory characterized by a unimodal and sharply skewed distribution that peaks near $m_i = 1$; in much the same way, Fig. 6b corresponds to a stable CW trajectory. Figure 6c exemplifies another type of pedestrian behaviour

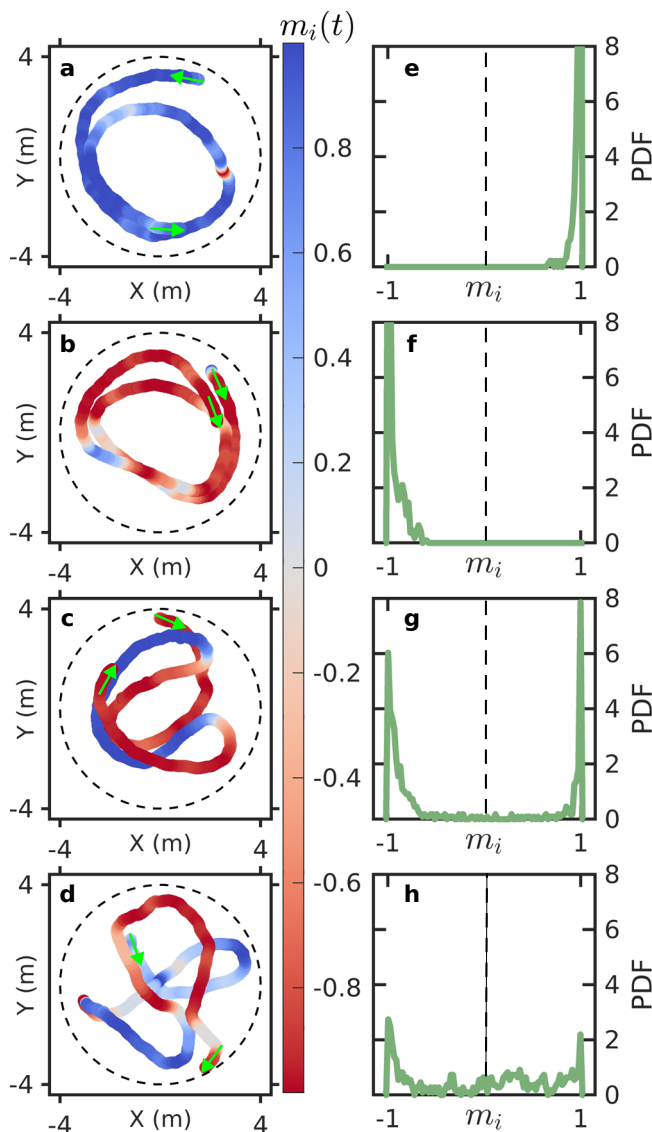


Fig. 6 | Four representative cases of individual rotational behaviour during confined motion in Japan. **a–d** Individual trajectories of four different pedestrians over 40 s coloured according to the instantaneous individual polarization value (m_i) (see colour bar on the right). Green arrows indicate the direction of motion at the start and end of each trajectory. Spatial units in both the vertical and horizontal directions are metres. **e–h** The corresponding probability density functions (PDFs) of m_i for each trajectory.

in which the rotating direction changes during the experiment (in this case, it changes twice). Accordingly, the PDF(m_i) shows a bimodal distribution with two marked peaks at $m_i = 1$ and $m_i = -1$. Finally, Fig. 6d shows a scenario in which the pedestrian rotates, but also performs a number of straight paths that give rise to more values of m_i different from ± 1 , and therefore to a broader distribution.

Considering the particularities of these distributions and aiming to reflect the individual behaviour using a single parameter, we computed the time-averaged individual polarization ($\bar{m}_i$) for each pedestrian. In this way, $\bar{m}_i \sim 1$ corresponds to pedestrians walking always CCW, $\bar{m}_i \sim -1$ corresponds to pedestrians walking always CW, while intermediate values (and particularly, those close to $\bar{m}_i \sim 0$) reflect both pedestrians that change rotating direction as in Fig. 6c and those performing straight trajectories as in Fig. 6d. In Fig. 7 we represent the distributions of $\bar{m}_i$ for all pedestrians that participated in each experiment (note that, for each experiment, we combined the results obtained in different conditions). Remarkably, in all cases the

distributions show a notable peak at $\bar{m}_i \sim 1$, revealing the presence of a number of people determinedly walking CCW, no matter the specific conditions at which the experiment was implemented. Also, the distributions suggest the existence of an analogous peak at $\bar{m}_i \sim -1$, but this is in general less prominent and altogether absent in the case of the nursery school experiments.

Overall, Fig. 7 indicates substantial individual variability in rotational behaviour. Despite this variability, in all experiments there is an important proportion of pedestrians exhibiting a determined preference for CCW rotation. Notably, this behaviour at the individual level helps to explain the main features reported for the collective polarization parameter M . In this way, the consistent positive values of M can be justified by the presence of a larger proportion of pedestrians moving CCW than CW. Similarly, the absence of a peak at $M = -1$ can be explained by the intrinsic variability of the pedestrian type of motion; with the exception of kids, in all cases there will be people walking CW or straight. Also, the findings reported in Fig. 7 suggest that the correlation between the sharpness of the PDF(M) and the crowd size is merely a statistical effect. When the crowd is small, each value of M is computed using a small number of values of m_i , and then the fluctuations increase just for statistical reasons.

Beyond this remarkable correlation between the macroscopic behaviour and the individual one, the results of Fig. 7 suggest that the prevalent preference for CCW rotation is not a collective effect but an individual one. Interestingly, this hypothesis is supported by the fact that the distribution with the sharpest peak at $m = 1$ occurs for the scenario in which pedestrians move with more freedom; i.e. the teenagers walking in a space free of boundaries (Fig. 7b).

Individual motion

Aiming to confirm that the CCW motion symmetry breaking is not caused by a collective effect but a result of individual preferences of motion, we implemented a new set of experiments in which over 200 participants walked alone (one at a time) in an enclosed arena (Fig. 8a). In these new tests, we looked for a connection between this hypothetical CCW motion preference of the individuals and some biological features, such as handedness, footedness, or eye dominance. To this end, each participant was asked about their dominant hand, foot, and eye (left or right). If they were unsure, dominance was determined through a series of performance tests (see *Methods* for details). Participants who showed no clear dominance (i.e., were ambidextrous or had indeterminate eye preference) were excluded from the analysis. Furthermore, 49 participants were asked to walk with a patch covering the right eye, a strategy that was aimed at evaluating whether artificially constraining the right visual field could have a significant effect on the rotational bias.

For each pedestrian, we extracted the complete trajectory within the arena (left panels of Fig. 8b) and obtained the instantaneous individual polarization $m_i(t)$. Then, we computed the probability density function of the polarization values of each individual. In the right panels of Fig. 8b, we show two examples corresponding to a pedestrian who is consistently walking CCW (top panels) and a pedestrian with several changes in the rotation direction (bottom panels). From these distributions, we calculated $\bar{m}_i$ for each pedestrian, and then built the distributions of PDF($\bar{m}_i$) (as in Fig. 7) by considering all participants, irrespective of their condition. Clearly, the distribution exhibits a pronounced peak near $\bar{m}_i \sim 1$, much higher than the one at $\bar{m}_i \sim -1$. This result provides evidence that the origin of the CCW motion is not at the crowd level, but at the individual one. Interestingly, the distribution also presents a peak for values of $\bar{m}_i$ slightly greater than 0 that is more prominent than for the individuals moving within a crowd (Fig. 7). We speculate that this might be related to psychological aspects as moving in an empty space with no other pedestrians might become unengaging, hence provoking the change in the rotation direction of pedestrians as in Fig. 8b bottom panels. Overall, the key

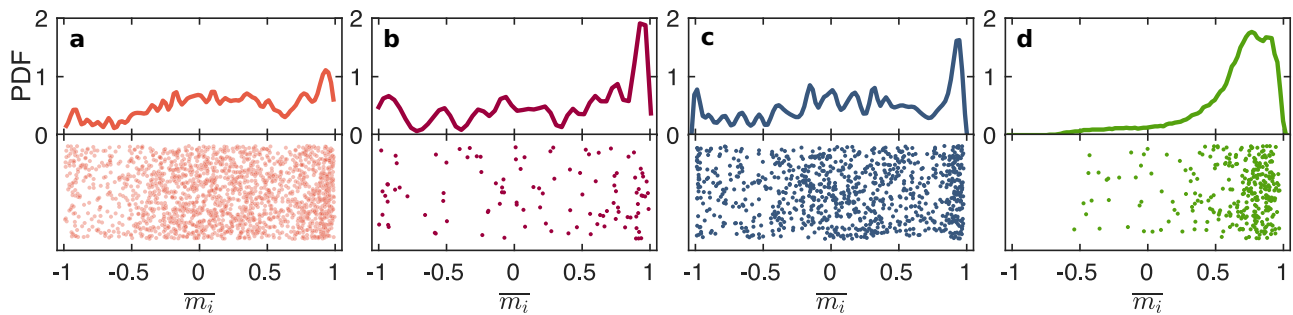


Fig. 7 | Average polarization values for each pedestrian in the four studied scenarios. Each panel includes a stripchart at the bottom, showing the time-averaged polarization values for each pedestrian ($\bar{m}_i$) and the corresponding

probability density function (PDF) of the whole set of these $\bar{m}_i$ values at the top. **a** Confined motion in Spain, **b** boundary-free motion of teenagers in Spain, **c** confined motion in Japan, and **d** kids motion in a Japanese nursery school.

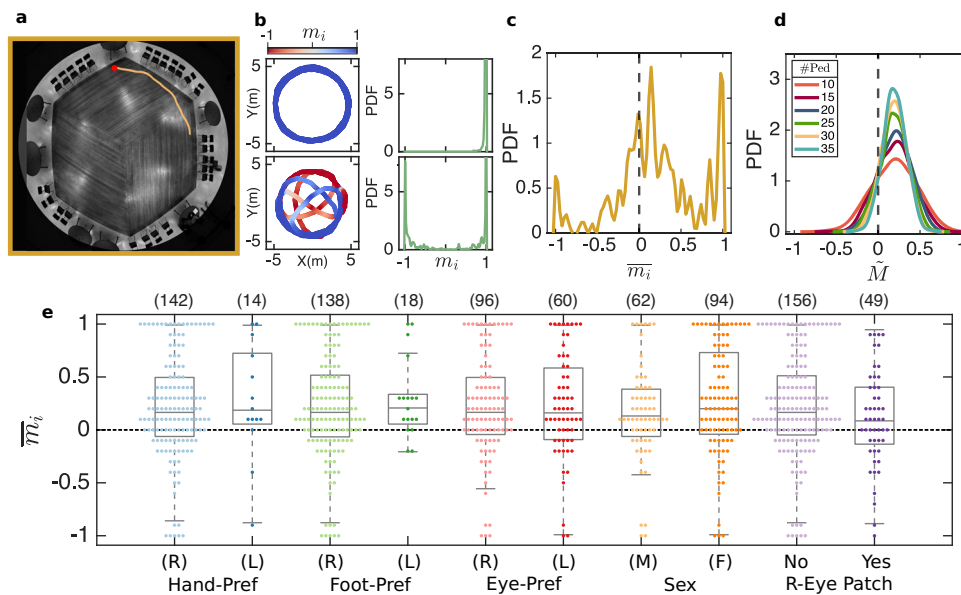


Fig. 8 | Rotational motion of individuals. **a** Snapshot of the last experimental setup, where individual pedestrians were instructed to walk alone and freely within an enclosure. The recent trajectory followed by the participant is shown in orange, with the current position marked in red. For image rights purposes, the pedestrian has been removed from this snapshot; only the arena background and the superimposed trajectory are displayed. **b** The trajectories of two pedestrians over a 60-s period are depicted, with colours representing their instantaneous individual polarization m_i , as indicated by the colour scale above. On the right, the probability density functions (PDFs) of m_i for each trajectory are shown. **c** Probability density function (PDF) of the individual time-averaged polarization values ($\bar{m}_i$). **d** Synthetic PDF of the collective polarization $\tilde{M}$, constructed by aggregating instantaneous individual polarization values m_i randomly selected from different pedestrians at

arbitrary times. Note that $\tilde{M}$ is not a genuine collective measure, but rather a synthetic construct designed to emulate its statistical properties. **e** Box plots of $\bar{m}_i$ of the data grouped by individual features: handedness preference (Hand-Pref), footedness preference (Foot-Pref), eye dominance (Eye-Pref), sex, and whether the right eye is patched or not (Right-Eye Patch). Note that the 'Yes' group corresponds to all participants wearing a patch, including both left- and right-eye dominant individuals. Each box depicts the median together with the interquartile range (IQR), while the whiskers extend to the most extreme values lying within 1.5 IQR. Outliers are not displayed; instead, each dot represents an individual data point associated with a pedestrian. The number in parentheses above each category indicates the sample size.

result of Fig. 8c is that the CCW asymmetry exists at the individual level. To statistically assess this bias, we performed a two-tailed one-sample Wilcoxon signed-rank test against 0: $z = -5.63$, $n = 156$, $P < 0.001$, $|r| = 0.45$. The median $\bar{m}_i$ was positive (bootstrap 95% CI for the median: 0.12–0.27), indicating that the median differs statistically from 0 and supporting the presence of a CCW bias at the individual level.

Next, we grouped the data according to pedestrian particularities such as handedness, footedness, eye dominance, and sex. Also, we discriminated the pedestrians who were asked to use a patch over their right eye. As shown in the box plots of $\bar{m}_i$ in Fig. 8e, the CCW bias remains consistent across all subgroups. Two-tailed Mann–Whitney U-tests showed no statistically significant differences in $\bar{m}_i$ between

right- and left-handed participants ($U(142, 14) = 898$, $z = -0.60$, $P = 0.554$, $r = -0.05$); the Hodges–Lehmann (HL) estimate of the location shift (Right – Left) was -0.05 with a bootstrap 95% CI of $[-0.34, 0.18]$. Analogously, no statistically significant differences were found between right- and left-footed participants ($U(138, 18) = 1134.5$, $z = -0.60$, $P = 0.553$, $r = -0.05$; HL Right – Left: -0.04 , 95% CI $[-0.21, 0.09]$), between right- and left-eyed participants ($U(96, 60) = 2851$, $z = -0.11$, $P = 0.917$, $r = -0.01$; HL Right – Left: 0.00 , 95% CI $[-0.15, 0.12]$), or between male and female participants ($U(62, 94) = 2542$, $z = -1.35$, $P = 0.178$, $r = -0.11$; HL Male – Female: -0.08 , 95% CI $[-0.24, 0.03]$). Likewise, the comparison between participants with and without a right-eye patch showed no statistically significant difference in $\bar{m}_i$ ($U(156, 49) = 4385$, $z = 1.55$, $P = 0.120$,

$r = 0.11$; HL No Patch – Patch: 0.10, 95% CI [−0.03, 0.25]). While these results indicate a consistent CCW bias regardless of these factors, we note that the sample sizes of certain subgroups—particularly left-handed ($n=14$) and left-footed ($n=18$) individuals—were relatively small, limiting statistical power. Therefore, although the trend is robust, we cannot entirely exclude the possibility of weak or subtle effects. Together, these results support the hypothesis that the CCW motion bias arises from individual locomotion trends rather than group-level phenomena. We found no statistically significant evidence that this intrinsic breaking of symmetry depends on the laterality-related biological features considered in this study.

Going a step further, we try to connect the individual motion results with the values of collective polarization (M mostly around 0.2 in Fig. 1) as well as their distributions, where we observed a clear narrowing of the peak with the crowd size (Figs. 2, 4). To this end, in Fig. 8d we synthetically build the PDFs of M by taking the actual values of m from the individual pedestrians moving alone (as shown in Fig. 8b) in a random manner. In particular, for each crowd size, we randomly select a subset of individuals and take one polarization measurement from each to calculate the hypothetical group-average polarization, $\tilde{M}$. We repeat this process for 1000 subsets, obtaining a statistically robust distribution of $\tilde{M}$ values as shown in Fig. 8d. Importantly, from data obtained for pedestrians walking alone, we obtain synthetic distributions of global polarization that peak at $\tilde{M} \sim 0.25$ and become systematically narrower as the crowd size increases; exactly as it happened with real experimental distributions of collective polarization (Figs. 2, 4). This finding corroborates the idea that the individual preferences of motion are likely the most important features observed at the collective level.

Discussion

In this work, we have implemented a series of experimental realizations—conducted in diverse conditions across Spain and Japan—that demonstrate the robust and widespread occurrence of CCW motion. Our findings are highly consistent: regardless of crowd size, boundary effects, or laterality traits such as handedness, footedness, and eye dominance, CCW motion systematically emerges. This reproducibility across varied settings, including two countries with different social norms and experiments with adults and children, supports the strength and reliability of our conclusions.

Traditionally, most of the emergent collective behaviours in pedestrian dynamics have been attributed to local interactions and social behaviour (including many different factors that are not mutually exclusive)^{35,36}. However, our data reveal that the CCW preference does not arise from these interactions. Instead, our results indicate that this symmetry-breaking phenomenon is fundamentally rooted in individual locomotor tendencies. This challenges the prevailing assumption that group-level behaviours are more than the sum of individual behaviours. We have even ruled out the possibility that hitherto unknown social norms could be the cause of the CCW motion. Therefore, our work exemplifies an instance of crowd motion which can be primarily explained without the need to resort to collective effects, nor to the specificities of pedestrian interactions (among them or with the environment). Moreover, the validation of our results through carefully controlled isolated trials suggests that intrinsic locomotor predispositions are a fundamental aspect of crowd behaviour.

The lack of an explanation for the origin of such individual biases opens new avenues for future research aimed at unravelling their biological or neurological grounds. Interestingly, preliminary studies of blindfolded walking show that some individuals spontaneously circle in one direction due to accumulating drift in their subjective straight-ahead, possibly linked to vestibular noise³⁷. While our experiments include full visual information, such findings suggest that

subtle sensorimotor asymmetries might also contribute to the CCW bias observed here. In this context, virtual reality experiments^{38,39} may provide a particularly powerful framework for studying this problem, as they allow sensory inputs to be isolated and systematically manipulated under controlled conditions.

At the same time, the generality of this behaviour should not be construed as evidence of a universal law. Our experiments reveal a robust underlying bias that tends to emerge when competing influences are minimal. In more complex scenarios—such as emergency evacuations, dense crowds, or environments shaped by signage or architectural constraints—other mechanisms may attenuate or completely eliminate the CCW tendency. For example, a recent study showed the emergence of chiral motion within a very dense crowd; however, the symmetry does not appear to be broken, as the size and number of CW and CCW regions have been shown to be similar⁹. In more diluted scenarios, one way to test the applicability limits of CCW motion is by considering tasks with strong symmetry and explicit goals. A first step in this direction is the analysis of antipodal (circle-antipode) experiments, where participants start from opposite points and converge towards a common centre before dispersing⁴⁰. Despite the imposed goal structure and high central symmetry, preliminary analysis of such experiments reveals that CCW bias is still present (see Supplementary Fig. 3). This finding suggests that the CCW preference is more resilient than anticipated, even under structured conditions. Moreover, our study participants were healthy children, teenagers, and young adults, leaving open the question of whether the same tendency holds in older adults or individuals with functional limitations, whose mobility, balance, and perception may differ substantially. Exploring these types of groups could not only test the limits of CCW motion but also provide valuable insights of its biomechanical origin.

Beyond pedestrians, it seems rather plausible that CCW or CW preference motion also occurs for other animals or biological systems. Indeed, vortex-like behaviours have been reported in fish schools, which alternate between schooling and milling states⁴¹, tadpole aggregations where circular motion enhances feeding efficiency⁴², and insect collectives such as army ants, whose mills can persist for days⁴³. Intriguingly, turning biases have also been documented at the individual level in some cases. For example, *Temnothorax* ants display a marked tendency to turn left while exploring⁴⁴, and flying budgerigars exhibit lateral preferences when choosing equivalent apertures during route choice⁴⁵. Taken together, these parallels suggest that rotational coordination is a widespread motif across living systems, and that the CCW bias in human crowds may represent a manifestation of a deeper biological principle of symmetry breaking.

Overall, the implications of our findings are significant. By demonstrating that individual biases—rather than collective effects—drive the observed CCW motion in pedestrian roaming, our study deepens our understanding of pedestrian dynamics and provides a new lens for studying crowd behaviour. These findings refine our theoretical framework while opening promising avenues for practical applications in high-traffic settings like airports, train stations, museums, and shopping centres. For instance, circulation paths in museums or exhibitions could be designed to follow the natural CCW bias to potentially improve pedestrian comfort and flow efficiency, while entry/exit points in large open spaces—such as plazas or stadium forecourts—might be more effective when aligned with this prevailing directional tendency. However, further research is needed to determine whether these individual tendencies persist in much more complex, real-world environments featuring static obstacles and varied pedestrian flows^{46,47}. Controlled investigations of these dynamics could ultimately enhance urban design and crowd management strategies, paving the way for innovative, people-centric public spaces.

Methods

Experiments

To investigate the underlying mechanisms contributing to the CCW bias observed in pedestrian dynamics, we conducted a series of six carefully designed experiments. Each experiment was tailored to test a specific hypothesis, ranging from the effects of physical interactions in confined spaces to the influence of social norms and individual-level characteristics. The experiments are presented in chronological order, reflecting the historical development of the project: initial studies adopted participant numbers consistent with previous related work¹⁶, while subsequent campaigns progressively adapted the sample sizes (both in terms of group size and number of repetitions) to address new hypotheses. Across all experiments, participants were healthy children, teenagers, or young adults, ensuring consistency across conditions while leaving the exploration of older groups or people with functional limitations for future research. All procedures were conducted in accordance with all relevant ethical regulations. The following subsections provide a detailed overview of the experimental setups, methodologies employed, participant characteristics, and procedures implemented in each study.

Confined random motion in Spain. The first experiment was conducted at The University of Navarra, Spain. The arena was a confined circular region of 5 metres radius, enclosed by 1-metre-high fences (Fig. 1a).

A total of 50 participants (23 men, 27 women; age 25.7 ± 4.08 years) were recruited through an online form. Informed consent was obtained from all participants prior to participation. The University of Navarra's ethics committee determined that its review was not needed, as the data were anonymised from the start, no personal information was collected, and no faces were recorded during the experiments.

Upon arrival, each participant completed a preliminary task involving a walk along a marked straight line towards a wall, turning at the wall, and returning to the starting point. This task allowed us to categorize participants as left-turners (LT) or right-turners (RT) based on the direction of their turn. Based on this classification, we constructed different experimental conditions varying the number of pedestrians inside the enclosure (global density) and distinct proportions of RT. A complete description of all experimental conditions can be found in Supplementary Table 1.

The sequence of each experiment was as follows:

1. **Initial Positioning:** Participants positioned themselves at pre-marked starting points (black crosses on the ground) looking towards one of the four coloured equidistant posts that were placed outside. Importantly, the post each pedestrian was asked to look at was randomly determined by an individual code provided to each participant on a card.
2. **First Random Phase:** Following a starting signal, participants started walking for 40 s. At the start of the experiment, they were instructed to move continuously without stopping and avoiding following others. The first 10 s of this phase were excluded from the analysis to avoid any possible transient effect.
3. **Target Motion:** Participants were instructed to move towards one of the four coloured posts near the fence, touch the fence close to the post, and then resume walking.
4. **Second Random Phase:** Random walking resumed for another 40 s, with the first 10 s excluded due to motion transitions following the target interaction task.
5. **Repeated Target Motion:** Participants were prompted to repeat the action of moving to the posts (different from the previous ones).
6. **Third Random Phase:** Random walking continued for 40 s, again excluding the initial 10 s.

In total, three random-walking intervals of approximately 30 s each were analysed per experiment. Each experimental condition was repeated twice, but with different individuals, resulting in six different phases of random motion to analyse.

All experiments were recorded using a camera mounted 10 metres above the ground, positioned directly above and pointing towards the centre of the arena. The footage was captured at a resolution of 4000×3000 pixels at 15 frames per second. A custom-built image analysis programme was used to track the positions of all pedestrians in the videos. To correct image deformation caused by the camera, the positions were calibrated using approximately 40 images of a reference chequerboard. Velocities were calculated over a sliding window of 0.8 s to ensure pedestrians had moved a sufficient distance, minimising spurious noise.

Teenagers in a Spanish schoolyard. The experiment was conducted at the Hijas de Jesús secondary school in Pamplona, Spain. The schoolyard, measuring approximately 50×60 m² (see Fig. 1b), constituted the experimental area. A total of 107 students, aged 13–14 years, participated in the study. Before conducting the experiment, the research team met with the school headmaster to explain in detail the experimental procedure, the type of data to be collected (video recordings without identifiable faces), and the anonymization process. Following this meeting, an information sheet describing the activity and its scientific objectives was distributed to all parents through the school's digital communication platform six days before the experiment. Parents were presented with a two-point authorization statement via the school's digital platform: (1) permission for their child to participate in the experiment, and (2) consent to the anonymized capture of images for strictly scientific purposes. They responded by clicking to authorize or decline directly within the platform, with a response deadline set three days before the experiment. Only students whose parents or legal guardians had authorized through the platform were permitted to participate; informed consent from parents or legal guardians was thereby obtained for all 107 participating minors. All collected data were anonymised from the outset, with no personal identifiers or facial images recorded. Under these conditions, The University of Navarra's ethics committee determined that no further approval was necessary.

In accordance with the methodology used in the aforementioned studies, participants were instructed to walk freely within the schoolyard, avoiding stopping or forming clusters. Initially, all students were gathered in a circular region marked on the ground. After the starting signal, the experiment began and lasted for 100 s. To avoid any potential bias due to initial positioning, the first 35 s of the experiment were excluded from the analysis. The remaining 65 s were used to evaluate the collective and individual rotational behaviours of the participants. The same experiment was repeated twice.

A DJI drone was used to capture footage from a height of 40 metres. The recordings were made at a resolution of 1920×1080 pixels with a frame rate of 30 fps. To stabilise the videos before performing the tracking, the point feature matching approach (from the OpenCV library) was applied. A custom-developed image analysis programme was employed to track pedestrian positions and velocities. To enhance accuracy and reduce noise, velocity calculations were performed using a sliding window of 0.8 s.

Confined random motion in Japan. The experiment was conducted at The University of Tokyo, Japan, within a circular enclosure with a radius of 4 metres, delimited by chairs. A total of 39 participants (25 men and 14 women, aged 26.8 ± 7.4 years) were recruited through a website⁴⁸. All participants provided informed consent, and the experimental protocol was approved by the Research Ethics Committee of the Graduate School of Engineering at The University of Tokyo.

Similar to the protocol implemented in Spain, participants were categorised as either left-turner or right-turner by asking them to perform a turn towards a wall prior to the experiments. An improvement introduced in this study was repeating this assessment two other times at different moments of the two-hour experiment: once during the first break and again at the conclusion. This allowed us to evaluate the consistency of turning direction among participants. A high level of consistency ($\sim 85\%$) was observed, with participants turning in the same direction in all three cases (see Supplementary Table II for details). The experimental conditions varied in terms of global density and the percentage of RT, as detailed in Supplementary Table I.

Each experimental trial lasted 45 s. Staff members first placed participants randomly throughout the enclosure to ensure a broad spatial distribution. Upon hearing a sound signal, they began walking randomly, as in the Spanish experiment. Another signal marked the end of the trial. To avoid transient effects, the first and last 5 s of each trial were excluded from the analysis. Each experimental condition was repeated four times, changing the participants included in the test whenever possible.

The experiments were recorded using a camera mounted 6 metres high in an azimuthal position above the centre of the room. The recordings had a resolution of 1920×1440 pixels and a frame rate of 30 fps. Videos were later analysed using PeTrack software^{49,50} to extract the coordinates of pedestrians based on their coloured caps. To facilitate identification, left-turners participants were given yellow hats, while right-turners participants were given red hats to wear during the experiments (discrimination was performed based on the turning direction in the initial test before the experiments). As in the Spanish experiments, velocities were calculated using a sliding window of 0.8 s.

Given that the experiment was conducted in Japan, where we hypothesized that lane formation predominantly occurs on the left side, it was essential to confirm whether this assumption held true. To validate this hypothesis, a questionnaire was administered to participants during the informative session for the experiments. The questionnaire presented participants with a series of images showing an individual walking in a corridor at varying distances (see Supplementary Note 3). Participants were asked to indicate their preferred walking direction. The results of this survey (summarized in Supplementary Table 2) revealed a clear preference for walking on the left side (Supplementary Note 1), confirming the hypothesis.

Nursery school in Japan. This experiment, conducted by Jun Ichikawa et al.¹⁷, analysed the emergence of spontaneous social movement in children during eurhythmic activities in a nursery classroom setting. Specifically, the study examined a warm-up activity where children ran freely around the room while the instructor played the piano. As described in the original work, no instructions were given regarding the direction, purpose, or manner of running. The environment was symmetric, and the teacher remained passive, neither guiding nor encouraging particular behaviours. As reported in the original study, the protocol was approved by the ethics and safety committees of Kyoto Institute of Technology, the University of Electro-Communications, and Tamagawa University, and informed consent was obtained from the nursery principal and the parents/guardians of participating children.

Data were collected from four distinct class groups, each with a different mean age, all under six years. The individual positions of the children were included as Supplementary Information in the article. Velocities were calculated from these positions using a 0.8-s sliding window, as the videos were recorded at 20 fps. For each home-room, several periods of motion were analysed, lasting between 5 and 10 s. These periods were interspersed with pauses when the instructor stopped playing. The number of periods per group varied (see Supplementary Table I), with at least two repetitions recorded for each class. For further details about the experimental setup and methodology, readers are referred to the original article¹⁷.

Elicitation of social norms. For our elicitation of social norms, we resorted to well-established concepts and methods proposed by Bicchieri^{32,34}. Participants were students from The University of Navarra, approximately half of them Spanish and half from different foreign countries. The procedure was notified to the Ethics Committee of The University of Navarra, which determined that no further review was required (no personal data were collected and only three non-sensitive questions were asked). Informed consent was obtained from all participants prior to their completion of the questionnaire. They were shown the picture in Fig. 5a and went through the following questionnaire:

1. This is a ring with people inside. Picture yourself as if you were one of these persons, and you would like to walk in circles. In which sense would you move? (CW/CCW)
2. In which direction do you think most of the people will move? (CW/CCW)
3. In which direction do you believe that others expect you will move? (read twice) (CW/CCW)

Question (1) elicits the decision the person would take in the hypothetical case. Questions (2) and (3) elicit empirical and normative expectations, respectively.

Answers to the questionnaire were economically incentivized in order to obtain more careful responses. To that end, answers to questions (2) and (3) were compared to experiments and with answers to question (1) respectively; people answering questions (2) and (3) correctly (meaning either in agreement with the experiments for question (2) or with the majority of answers in question (1) for question (3)) entered a lottery for four gift cards of 25 euros each. We received a total of 168 valid responses.

Individual motion. The last experiment, conducted at The University of Navarra, aimed to examine the influence of individual characteristics on the turning preference of pedestrians. A total of 209 participants (88 men, 121 women) were recruited over three days. All participants provided verbal consent because recruitment was spontaneous: participants were adult university members approached as they passed near the experimental area. Prior to participation, each individual was verbally informed that the study was minimal risk and involved 60 s of free walking, with no facial images or personal data collected beyond the experimental characteristics reported. This verbal consent procedure and the spontaneous recruitment approach were communicated to the University of Navarra's ethics committee, which waived further ethics oversight.

The experiment was conducted within a hexagonal enclosure delimited by chairs and tables, with each side measuring approximately 4.6 m (Fig. 8a). Prior to the experiment, all participants underwent a series of assessments to determine their hand, foot, and eye dominance (left or right preference). For those who were unaware of this information, various tests were conducted. Foot dominance was determined by instructing participants to kick a wooden object a couple of times; if they alternated between feet, they were subsequently asked to simulate stepping on an insect to identify a consistent preference. Eye dominance was evaluated using the Miles test⁵¹. Independently, to examine how reducing the visual field affects movement patterns, 49 participants wore a patch over their right eye. The patched group included both left- and right-eye dominant individuals, and was compared against the rest of the participants who walked without a patch.

During the experimental trials, individuals were instructed to move freely within the enclosed space for 60 s without stopping. The experiments were recorded using a GoPro camera mounted 5 metres above the centre of the arena, capturing footage at a resolution of 3840×3360 pixels and a frame rate of 25 fps. The videos were analysed using a custom-developed image analysis programme to track the positions. As always, velocities of the participants were calculated using a sliding window of 0.8 s.

Metrics and nomenclature

For all experiments, the rotation (both individual and collective) was quantified by means of the polarization parameter¹⁸. Given a pedestrian i at time t , the individual polarization value m_i is defined as:

$$m_i(t) = \hat{v}_i(t) \cdot \hat{e}_i^\varphi(t) \quad (1)$$

where $\hat{v}_i$ is the normalized velocity vector, and $\hat{e}_i^\varphi$ is the azimuthal position calculated as $\hat{e}_i^\varphi = R_{90^\circ} \hat{r}_i$. This represents a 90° CCW rotation of the normalized position vector of the pedestrian $\hat{r}_i$ relative to the centre of the arena. In experiments conducted in confined circular arenas, the reference point is naturally taken as the geometrical centre of the arena. For the schoolyard experiments, where no fixed central point exists, the reference centre used for computing polarization was the global centre of mass of the crowd, averaged over the entire sequence. The stability of this choice was confirmed by analysing the position of the instantaneous CM, which remained within ~1 m of the global CM throughout the trial (see Supplementary Fig. 1). Thus, when $m_i = 1$, it indicates a perfect CCW circular motion, while $m_i = -1$ correlates with CW rotation.

Based on this, we can define the following magnitudes:

- $\bar{m}_i$: Time average of the individual polarization per pedestrian over all the experiment duration.
- Collective polarization $\mathbf{M}(t)$, defined as:

$$M(t) = \frac{\sum_i^N m_i(t)}{N}$$

where N is the total number of people in the arena.

- $\bar{\mathbf{M}}$: Time average of the collective polarization over all the experiment duration.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

All data generated in this study have been deposited in the Zenodo repository at <https://doi.org/10.5281/zenodo.19592341>. The deposited dataset includes the raw trajectories of all participants across all experiments, as well as the data supporting the plots presented in this paper. This study also makes use of a previously published dataset: the nursery-school trajectories reported, made publicly available by Ichikawa et al.¹⁷, who gave us written permission to use the data.

Code availability

The code used to conduct the statistical analyses and to generate the figures is deposited in the Zenodo repository⁵² at <https://doi.org/10.5281/zenodo.19592341>.

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Acknowledgements

We thank L. Urrea and C. Martín-Gómez for their help with the experiments. We also thank D. Maza for useful discussions and appreciate the

support from www.jikken-baito.com in recruiting participants in Japan. In addition, the authors would like to thank Jun Ichikawa for openly sharing experimental trajectories from the nursery school and for valuable discussions. We are also grateful to Ezel Üsten for insightful discussions on the social aspects of this work.

Author contributions

I.E.H, A.G. and I.Z. designed the research. I.E.H, C.F., A.G. and I.Z. designed the experiments. I.E.H, C.F., Z.S., A.G. and I.Z. built the experimental setup and recorded the experiments. A.S. designed the social norm elicitation section. I.E.H, Z.S., A.S., A.G. and I.Z. analysed the data. I.E.H., C.F., K.N., A.S., A.G. and I.Z. wrote the paper. All authors revised the results and commented on the manuscript.

Funding

I.E.H, A.G. and I.Z. acknowledge support from the Spanish Ministry of Science and Innovation through the Grants No. PID2020114839GB-I00 and No. PID2023-146422NB-I00 funded by MICIU/AEI/10.13039/501100011033, FEDER, UE. C.F. acknowledges support from the JSPS KAKENHI Grant No. JP23K13521 and JP26K07940, and K.N. acknowledges support from the JST-Mirai Programme Grant Number JPMJMI20D1. A.S. acknowledges support from grant PID2022-141802NB-I00 (BASIC) funded by MCIN/AEI/10.13039/501100011033 and by ‘ERDF way of making Europe’, and also from grant MapCDPerNets—Programa Fundamentos de la Fundación BBVA 2022.

Competing interests

The authors declare no competing interests.

Additional information

Supplementary information The online version contains supplementary material available at <https://doi.org/10.1038/s41467-026-73713-w>.

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Peer review information *Nature Communications* thanks Karol Bacik, Enrico Ronchi, Yao Xiao, and the other, anonymous, reviewer(s) for their contribution to the peer review of this work. A peer review file is available.

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